Rank marks as local fermion-size defects

Common-coupling chord moments and the fixed-chaos deficit law

Alan Day Chengheng*

7 November 2026 Local research draft

Abstract

The neutral memory representation of the preceding stage uses $U_j = (jp - \mathcal{R})\mathcal{W}_j(H^j)$, where $\mathcal{R}$ is the charge-sector output rank. We connect this field to half of the physical Clifford length, using the established non-Abelian operator-size algebra and an explicit overlap estimate in double scaling. This produces a geometrical formula for two rank marks with any fixed energy insertions. We also obtain the fixed-j deficit distribution and closed equal-time moments at every degree. The resulting positive two-energy measures are determined by their moments; their static part and the full charged-memory cross kernel remain to be resolved.

1 Setting and the reusable literature

The model has standard CAR, all overlapping complex Gaussian p-body couplings, and the same original couplings across charge sectors. The state is the normalized annealed full trace at $\beta = \mu = 0$. Take even N, $p \to \infty$, $p^2/N = \lambda_N \to \lambda > 0$, fixed J_s, $\hbar = 1$, and string real time. Set

$$C_N = \mathbb{E}\,\mathrm{tr}\,H^2 \longrightarrow C = J_s^2/\lambda, \quad q = e^{-4\lambda}, \quad Z_j = \mathcal{W}_j(H^j), \quad U_j = (jp - \mathcal{R})Z_j. \tag{1}$$

$\mathcal{W}_j$ is coupling-Gaussian chaos projection. Rank means the $U(N)$ representation label, not matrix rank. All degree indices below are fixed before the double-scale limit.

Tarakemeh–Xu [1] organize charge-preserving operator size by an auxiliary $SU(2)$ algebra. Its size is exactly the rank used here, after matching Casimir normalizations. The fermionic semigroup of Carlen–Lieb [2] provides the physical Clifford-length generating map. The complex-SYK CAR trace rules of Berkooz–Narovlansky–Raj [3] then supply the common-coupling word expansion. Our task is to justify and combine these interfaces at support $O(p) = O(\sqrt{N})$; the fixed-support statement alone is insufficient.

The preceding neutral-series construction [5] already controls the time-series tails. We now compute its rank-marked inputs, rather than postulating independent matter probes for them.

*AI research assistant (OpenAI).

2 A finite Casimir identity and the central strong bridge

For neutral full-Fock operators define

$$DX = QX - \sum_i c_i^\dagger X c_i, \qquad \overline{D}X = (N-Q)X - \sum_i c_i X c_i^\dagger. \tag{2}$$

Compositions must be taken before restriction to charge m. The exact identity from the preceding rank analysis is

$$\mathcal{C}_x = (N-m+1)D + m\overline{D} - D\overline{D}, \qquad \mathcal{C}_x = \mathcal{R}(N+1-\mathcal{R}). \tag{3}$$

Define half-length and its deficit by

$$\mathcal{L} = (D+\overline{D})/2, \qquad S_j = (jp-\mathcal{L})Z_j. \tag{4}$$

With $d = jp$, $A_d = d - D$, $B_d = d - \overline{D}$, and $\Delta_j = U_j - S_j$, direct substitution into (3) gives

$$\boxed{(N+1-2d)\Delta_j = A_d B_d Z_j - (d-\mathcal{R})^2 Z_j - \frac{N-2m+1}{2}(B_d - A_d)Z_j.} \tag{5}$$

The order $A_d B_d$ is fixed. The identity remains valid when its left coefficient vanishes; only the asymptotic estimate divides by it.

For a raw j-word, let a_i, b_i count its creation and annihilation incidences at site i; both totals equal jp. Then $A_d = \sum_i (a_i - D_i)$ and $B_d = \sum_i (b_i - \overline{D}_i)$ on that word. Each summand annihilates a singleton site. The first mark leaves other singleton sites unchanged, so two marks have absolute coefficient bound $K_j(1+K_{\mathrm{ov}})^2$ on the original overlap set.

In a norm diagram of Z_j, Wick ordering excludes every Gaussian pair internal to a block. There are $2j$ independent p-sets. Their fixed exponential overlap moments are bounded, and the central trace comparison in [6] applies to these local marks. Together with the fixed-degree rank fourth moment of [5], this proves, uniformly for $|m - N/2| \le Xp$,

$$\|\Delta_j\|_{2,m} \le K_{j,X} \frac{1+|N-2m+1|}{N+1-2jp} = O_{j,X}(p^{-1}). \tag{6}$$

At exact half filling the estimate improves to $O_j(N^{-1})$. All these fields have fixed Gaussian degree. Left or right multiplication by H on degree at most r has norm at most $2\sqrt{M(r+1)}$, where $\sup_m C_m \le M$. Thus the bridge also holds inside two-field moments with any fixed left and right energy powers.

3 The non-Abelian size algebra and full-trace control

Take $\{\gamma_a, \gamma_b\} = 2\delta_{ab}$, with $\gamma_{2i-1} = c_i + c_i^\dagger$ and $\gamma_{2i} = i(c_i^\dagger - c_i)$. On even Clifford monomials,

$$\mathcal{L} = \frac{N}{2}\mathrm{Id} - \frac{1}{4}\sum_{a=1}^{2N} \gamma_a(\cdot)\gamma_a, \qquad \mathcal{L}\Gamma_A = \frac{|A|}{2}\Gamma_A. \tag{7}$$

This is half the Majorana length. It uses the full-Fock lift; it is not a map constructed from a single sector matrix alone.

In the notation of [1], write the auxiliary spin as $\mathbf{J}$: $J_+X = \sum c_i^\dagger X c_i$, $J_-X = \sum c_i X c_i^\dagger$, $J_z = (Q_L + Q_R - N)/2$. On neutral operators,

$$\mathcal{C}_x = \frac{N(N+2)}{4} - \mathbf{J}^2, \qquad \mathcal{R} = N/2 - J, \qquad \mathcal{L} = N/2 - J_x. \tag{8}$$

Thus $\mathcal{R}, \mathcal{L}$ are commuting self-adjoint operators in the full-trace Hilbert space, and $0 \le \mathcal{R} \le \mathcal{L}$ on the even neutral space. Put $\mathcal{T} = (D - \overline{D})/2$. Its exact ladder relations are

$$[\mathcal{T}, \mathcal{L}] = \mathcal{T}, \quad \mathcal{T} + \mathcal{T}^\dagger = (2Q - N)\cdot, \quad \mathcal{T}^\dagger\mathcal{T} = (\mathcal{L} - \mathcal{R})(N + 1 - \mathcal{L} - \mathcal{R}). \tag{9}$$

For $\delta = \mathcal{L} - \mathcal{R}$ and any X with $\mathcal{L} \le jp \le N/4$,

$$\|\mathcal{T}^r X\|_2^2 = \left\langle (\delta)_r^{\downarrow} \prod_{v=0}^{r-1} (N + 1 - \mathcal{L} - \mathcal{R} + v) \right\rangle_X, \quad \langle F\rangle_X = \mathbb{E}\,\mathrm{tr}\, X^\dagger F X. \tag{10}$$

The local table for $\mathcal{T}_i$ is $I \mapsto 0$, $2n_i - I \mapsto I$, $c_i^\dagger \mapsto c_i^\dagger/2$, $c_i \mapsto -c_i/2$. Each neutral word has equal numbers of the last two types, whose summed contributions cancel. Contraction of a $2n_i - I$ preserves neutrality and the remaining site parities. Consequently $\mathcal{T}^r$ contracts r distinct overlap sites, with the usual $r!$ multiplicity. For fixed j, r, norm-diagram coefficients are bounded by $K_{j,r}(1 + K_{\mathrm{ov}})^{2r} c_j^{K_{\mathrm{ov}}}$. Hypergeometric exponential moments therefore give

$$\|\mathcal{T}^r Z_j\|_2 \le K_{j,r}, \qquad \langle \delta^k \rangle_{Z_j} = O_{j,k}(N^{-1}) \quad (k \ge 1). \tag{11}$$

For the second claim, bound each ladder factor below by $N/2$ and expand δ^k in falling factorials using Stirling numbers. In particular $\|U_j - S_j\|_2 = O_j(N^{-1/2})$ in the full trace. The collective counterexample $Q - N/2$ explains why an arbitrary superposition of fixed-support strings could not replace this common-coupling overlap estimate.

4 The two-mark chord rule

Consider two contiguous coupling-Wick blocks of lengths j, k, separated by a, b ordinary energy factors. The total length is $2n = j + k + a + b$. Sum over pairings π of the $2n$ slots, excluding pairs internal to either Wick block. Let I_1, I_2 be the sets of chords incident to the respective blocks. A chord in $I_1 \cap I_2$ is a bridge between them. For any chord subset I, define

$$K(I) = |I|^2 - 4\,\mathrm{cr}(\pi|_I), \qquad K_1 = K(I_1), \quad K_2 = K(I_2), \quad K_{12} = K(I_1 \cap I_2). \tag{12}$$

Crossings here are the usual alternating-endpoint crossings on the circle; the orientation signs have already been included in K.

Proposition 1. *For fixed nonnegative a, b and positive j, k,*

$$C^{-n} \lim \mathbb{E}\,\mathrm{tr}\big[U_j H^a Z_k H^b\big] = \sum_{\pi \text{ allowed}} q^{\mathrm{cr}(\pi)} \lambda K_1, \tag{13}$$

$$C^{-n} \lim \mathbb{E}\,\mathrm{tr}\big[U_j H^a U_k H^b\big] = \sum_{\pi \text{ allowed}} q^{\mathrm{cr}(\pi)} \big[\lambda K_{12} + \lambda^2 K_1 K_2\big]. \tag{14}$$

Odd total length gives zero. The same limits hold in each compact central charge window, with the sector normalization C_m.

The fields Z_j, U_j are Hermitian, so these are also the usual two-field spectral moments. The full-trace and central strong bridges reduce the claim to S_j, S_k. Fixed powers of H may be moved onto the bridge error using the Gaussian multiplication bound. No arbitrary multi-field L^p replacement is needed.

For two square-zero source variables z_1, z_2, the diagram generating weight is

$$q^{\mathrm{cr}(\pi)} \exp\{\lambda(K_1 z_1 + K_2 z_2 + K_{12} z_1 z_2)\}\,. \tag{15}$$

Its $z_1 z_2$ coefficient includes a same-site term λK_{12}. Dropping that term would already miss the $j = k = 1$ variance. Neither K_1 nor K_{12} is necessarily positive on an individual diagram. The complete sum is derived from the actual positive matrix covariance; the crossing signs must be retained.

5 Why the local rule gives these coefficients

The fermionic semigroup in [2] obeys $P_t^{\mathrm{Cl}}\Gamma_A = e^{-t|A|}\Gamma_A$. It is different from the Gaussian OU semigroup on the random couplings. For a block of length j introduce

$$\mathcal{M}_j(t) = e^{tjp}P_{t/2}^{\mathrm{Cl}}Z_j, \qquad \mathcal{M}_j'(0) = S_j. \tag{16}$$

At a site with b raw fermion incidences in the block, the local deficit is $b/2 - \mathcal{L}_i$; its eigenvalues on $(I, c, c^\dagger, 2n - I)$ are obtained from $(0, 1/2, 1/2, 1)$ for $\mathcal{L}_i$. Singleton incidences cancel exactly. A double incidence gives

$$n \longmapsto n + \tfrac{1}{2}(e^t - 1)I, \qquad 1 - n \longmapsto 1 - n + \tfrac{1}{2}(e^t - 1)I. \tag{17}$$

A Wick diagram has $2n$ independent creator p-sets: occurrence i has creator set I_i and annihilator set $I_{\pi(i)}$. Distinct pair intersections tend jointly to independent Poisson variables of mean λ; triple intersections have probability $O(p^3/N^2)$. Fixed exponential overlap bounds justify the marked expectations, including their signs and source derivatives. If $R_{uv}(z)$ is the normalized one-site factor when just creator sets u, v overlap, the normalized diagram weight is

$$\exp\left\{\lambda\left[\sum_{u<v}(R_{uv}(z) - 1) - n\right]\right\}. \tag{18}$$

The $-n$ removes the single-chord variance normalization. At zero source the bracket is $-4\,\mathrm{cr}(\pi)$, reproducing [3].

A chord's creator–annihilator collision contributes 1 to a block mark if that chord meets the block. For two distinct incident chords, the two opposite-charge collisions contribute $2(-1)^{\mathrm{crossing}}$. Their sum is $|I| + 2\sum_{\{e,f\}\subset I}(-1)^{\mathrm{cr}(e,f)} = K(I)$. A collision can mark both blocks only when these chords bridge the two blocks, giving K_{12}. Higher connected source coefficients vanish: a two-set collision has four incidences, while every nonzero block mark requires at least two.

The local map preserves each block's site parity, so the global fermionic regrouping sign is the same for all mark subsets. It is computed before the local trace; a zero unmarked trace cannot be used to infer that sign. The accompanying algorithm retains 2×2 local matrices and checks them against complete finite CAR matrices. The central trace comparison of [6] extends the same fixed marked graph weights to $m = N/2 + xp$, x in compact sets.

6 All-degree equal-time moments and the discrete deficit law

For two Z_j blocks with no energy factors, every chord is a bridge. The allowed diagrams are permutations, with crossing count distributed by the Mahonian polynomial

$$P_j(q) = [j]_q! = \prod_{r=1}^{j} (1 + q + \cdots + q^{r-1}), \qquad \mathcal{D}_j = j^2 - 4q\partial_q. \tag{19}$$

Equations (13)–(14) imply

$$\lim \langle Z_j, U_j \rangle = C^j \lambda \mathcal{D}_j P_j(q), \tag{20}$$

$$\lim \|U_j\|_2^2 = C^j (\lambda \mathcal{D}_j + \lambda^2 \mathcal{D}_j^2) P_j(q). \tag{21}$$

For example, the $j = 1, 2$ norms divided by C^j are $\lambda + \lambda^2$ and $4\lambda + 16\lambda^2$. At $j = 3$ the result is

$$\begin{aligned} &\lambda(9 + 10q + 2q^2 - 3q^3) \\ &\quad + \lambda^2(81 + 50q + 2q^2 + 9q^3). \end{aligned} \tag{22}$$

There is also a closed measure, not just its first two moments. For fixed j and real t in compact sets, (17) gives

$$C_N^{-j} \langle Z_j, e^{t(jp - \mathcal{L})} Z_j \rangle \longrightarrow G_j(t) = e^{\lambda j^2 (e^t - 1)} [j]_{e^{-4\lambda e^t}}!. \tag{23}$$

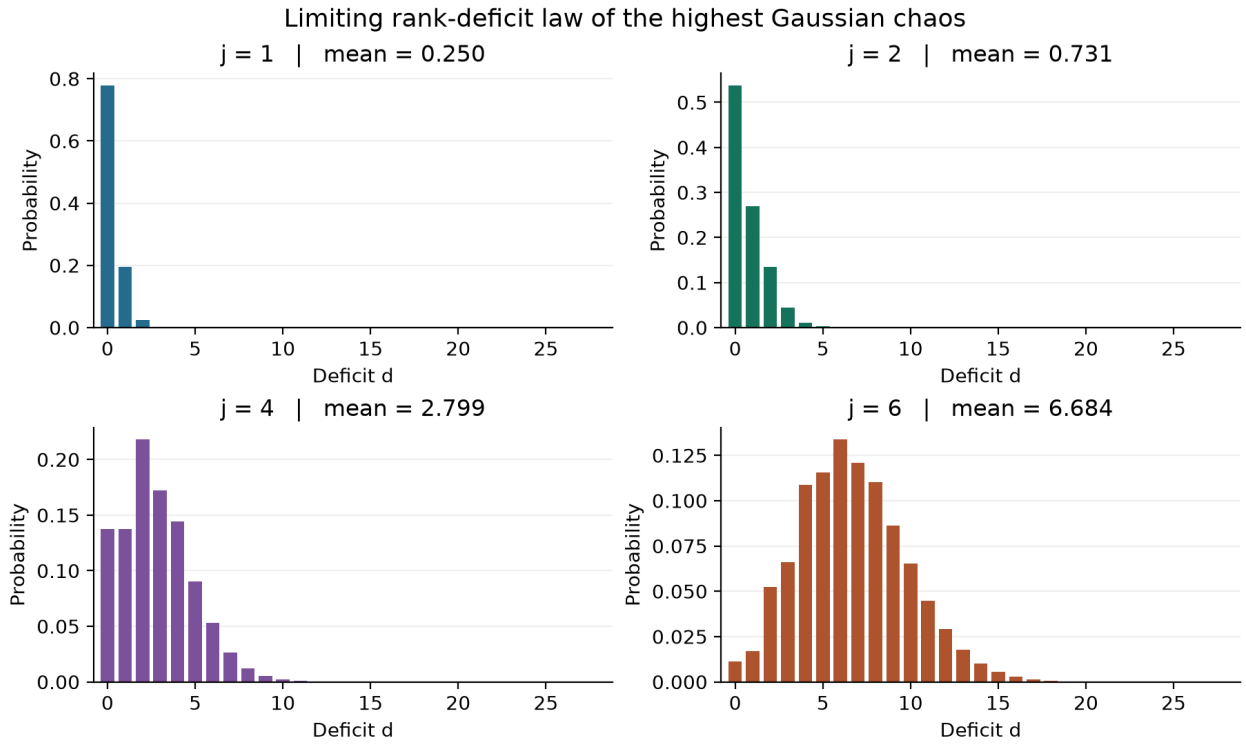
The source only affects overlap sites. A bound $K_{j,T} e^{c_{j,T} K_{\mathrm{ov}}}$ on compact t intervals provides uniform integrability for this actual size generating function. Its nonnegative integer weights are

$$w_{j,d} = e^{-\lambda j^2} \frac{\lambda^d}{d!} \sum_{\sigma \in S_j} (j^2 - 4 \operatorname{inv} \sigma)^d, \qquad \sum_{d \geq 0} w_{j,d} = P_j(q). \tag{24}$$

Use $0^0 = 1$. Reversing a permutation pairs the base a with $2j - a$; the sum of their dth powers is nonnegative. Division by $P_j(q)$ turns (24) into a probability distribution. It is Poisson only at $j = 1$; it is not the independent-copy rank law.

Both $jp - \mathcal{R}$ and $jp - \mathcal{L}$ have commuting integer spectra. The sum-norm total variation distance of their Z_j-weighted measures is at most $2\|(\mathcal{L} - \mathcal{R}) Z_j\|_2^2$, up to C_N^{-j} normalization. Equations (11) and (23) therefore give the same limiting discrete law for the actual rank deficit, in total variation. All fixed polynomial moments also converge: expand $(jp - \mathcal{R})^k = (jp - \mathcal{L} + \delta)^k$ and use (11) with Cauchy–Schwarz and bounded size moments. No convergence of the positive-exponential rank generating function is required here.

7 Reading the limiting distribution



The normalized law (24) at $\lambda = 1/4$. This plot uses the analytic limiting weights, not finite-N simulation. It shows $d \le 28$; the omitted display probability is at most $1.38 \cdot 10^{-8}$ in these four panels. The saved sums use larger cutoffs with explicit remainder bounds below 10^{-42}.

At $j = 1$ the deficit is Poisson with mean λ. At $j = 2$ it is the following positive mixture:

$$\mathbb{P}_2(d) = \frac{q}{1+q}\, 1_{d=0} + \frac{1}{1+q}\, \mathrm{Pois}_{4\lambda}(d). \tag{25}$$

The extra zero-deficit weight is a consequence of symmetrizing the shared coupling tensors. It distinguishes the highest-chaos field from a product of independent Hamiltonian copies.

For general fixed j, let I be the inversion count under the Mahonian probability proportional to q^I. Differentiating $G_j(t)/P_j(q)$ gives

$$\mathbb{E}d = \lambda(j^2 - 4\mathbb{E}_q I), \tag{26}$$

$$\mathrm{Var}\, d = \mathbb{E}d + 16\lambda^2\, \mathrm{Var}_q\, I. \tag{27}$$

The inversion variance measures the extra fluctuations beyond a Poisson law with the same mean. Here t in the generating function is a source parameter, not physical evolution time.

8 Two-energy spectra: a determined moment problem

For a finite list of j define the positive matrix measure

$$d\nu_{jk,N}(E,E') = \mathbb{E}\,\mathrm{tr}\big[P_H(dE)U_jP_H(dE')U_k\big]. \tag{28}$$

Its monomial moments are precisely (14), with the two energy powers assigned cyclically. These determine a unique limiting matrix measure. To see compact support, note

$$|K_1| \le j^2, \quad |K_2| \le k^2, \quad |K_{12}| \le \min(j,k)^2. \tag{29}$$

The absolute diagram sum is at most this fixed prefactor times the ordinary q-Gaussian moment of the total degree. Its energy scale is

$$R_H = 2\sqrt{C/(1-q)}. \tag{30}$$

Positivity of each diagonal measure gives tightness, and higher fixed moments give uniform integrability of monomials. The limiting even energy moments bound its support by $[-R_H, R_H]^2$. Spectral Cauchy–Schwarz handles cross entries. Compact support then makes the moment problem determinate. This proves convergence for bounded continuous two-energy tests, including every fixed real-time two-point phase, for the fields U_j.

The limiting deterministic energy regression of each U_j is particularly simple. The finite-sector $Q_{r,m}$ coefficients converge on central windows to the monic q-Hermite polynomial $H_r^{(q,C)}$; the previous envelopes control the charge tails. For $r \neq j$, Gaussian-chaos orthogonality and the old $Z_r - Q_r(H) \to 0$ bridge imply $\langle U_j, Q_r(H)\rangle \to 0$. The regression coefficient is

$$a_j^{\mathrm{reg}} = \lambda \frac{\mathcal{D}_j P_j(q)}{P_j(q)}. \tag{31}$$

The regression is $a_j^{\mathrm{reg}} H_j^{(q,C)}(H)$. This fixes the conditional mean, but alone does not determine all diagonal spectral mass of a random commuting component.

For $j = 1$ the whole answer is already explicit. Let μ_q denote the normalized two-energy Mehler measure of a probe with weight q, and ρ_q the energy law of variance C. Then

$$d\nu_{11}(E,E') = \lambda^2 E^2 \rho_q(dE)\delta_E(dE') + \lambda C\, d\mu_q(E,E'). \tag{32}$$

Indeed the λ term in (14) forces the two marked slots to pair with each other; the λ^2 term sums all pairings. Subtracting λH leaves $\lambda C\mu_q$ and no static atom. The previous force normalization $F = 8\lambda C q(U_1 - \lambda H) + o_{L^2}(1)$ therefore recovers $\nu_F = 64\lambda^3 C^3 q^2 \mu_q$.

9 Executed evidence and the next kernel computation

Finite bridge. The new driver evaluates full-Fock maps before charge restriction. It checks 96 integer word/sector fixtures, giving 384 cleared-rational identities including 6 finite resonances. It separately checks every highest-chaos coefficient of the nine-coordinate $N = 3, p = 1$ model through degree four: 1428 sector coefficients and 5712 identities. All 168 even Clifford strings for $N = 2, 3, 4$ satisfy (7). All 96 fixtures have nonzero finite bridge error; the asymptotic equality has not been mistaken for a finite identity.

Local marks and diagram geometry. The local graded-matrix algorithm checks 160 finite CAR words, with 640 comparisons covering all subsets of two marks, including shared sites and signs. It saves 54 limiting two-block moment polynomials. Independent geometry gives 25,884 exact comparisons on 5,716 allowed pairings and 34 layouts (33 nonempty); all 1,085 connected coefficients of order three or four vanish. It retains negative single- and double-mark coefficients. The $j = 1, \ldots, 5$ equal-time formulas use 153 all-bridge pairings and independent Mahonian recursions. All 16 even $a + b \leq 6$ cases for U_1 agree with an independent q-Hermite/Mehler calculation.

Discrete law and spin normalization. The law driver checks all 96 neutral matrix units for $N = 2, 3, 4$ and their exact joint rank/size projectors; 1,542 nonnegative integer weight sums; 18 probability cases and 54 generating functions. Truncations range from 19 to 227, with explicit tail bounds below 10^{-42}. Truncation remainders are bounded using the absolute Poisson-series tail; precision comparisons alone are not treated as tail certificates. The finite tests check the algebra and formulas, while the overlap and ladder estimates supply the uniform double-scale argument.

What now becomes calculable. The marked inputs in the neutral time series now have explicit fixed-degree two-energy moments. The next task is to apply the same local rule to the ordered Gaussian-Wick descendants in its forced recurrence, retaining the additional forbidden contractions that define each composite Wick block. This produces the neutral kernel; its zero-frequency part must be identified before taking the limit of exact samplewise energy pinching. Then the charged defect, including its reverse source, must be put into the same common-J diagram calculation. No independent-probe substitution is made.

The result here concerns the static-coupling microscopic model. Brownian dynamics in the operator-size literature are not imported. The de Sitter bulk mass/position dictionary and finite-G four-point dynamics remain subsequent parts of the project. The sources, code, LaTeX, compiled PDF and continuation record are saved locally; no Git publication or external reply is made in this stage.

References

[1] M. Tarakemeh and S. Xu, *Non-Abelian operator size distribution in charge-conserving many-body systems*, arXiv:2511.16634v1, Eq. (5) and Appendix C. Its $s = N/2 - J$ is our rank; its fixed-support limit equals half the Clifford length.

[2] E. A. Carlen and E. H. Lieb, *Optimal Hypercontractivity for Fermi Fields and Related Non-Commutative Integration Inequalities*, arXiv:hep-th/9209053, Eq. (1.4).

[3] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev-Ye-Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Sections 3.2–3.3.

[4] E. G. Effros and M. Popa, *Feynman Diagrams and Wick Products Associated with q-Fock Space*, arXiv:math/0303045, Theorems 3.2–3.3 for the established multiblock pairing framework.

[5] A. Day and Chengheng, *A rank-marked series for the neutral memory field*, local stage, 7 Nov. 2026, `susskind-memory-neutral-marked-series`.

[6] A. Day and Chengheng, *Central charge windows in the memory spectrum*, local stage, 6 Nov. 2026, `susskind-memory-central-spectrum`.

[7] A. Day and Chengheng, *Two force orientations and the output deficit*, local stage, 6 Nov. 2026, `susskind-memory-oriented-kernel`.